

```

31.000 1081507.453 8592.286 33167.510 41759.797
32.000 1048084.276 8336.620 33423.177 41759.797
33.000 1014403.463 8078.983 33680.814 41759.797
34.000 980463.026 7819.360 33940.437 41759.797
35.000 946260.965 7557.736 34202.061 41759.797
36.000 911795.264 7294.095 34465.702 41759.797
37.000 877063.889 7028.422 34731.375 41759.797
38.000 842064.793 6760.701 34999.096 41759.797
39.000 806795.913 6490.916 35268.880 41759.797
40.000 771255.168 6219.052 35540.745 41759.797
41.000 735440.463 5945.092 35814.705 41759.797
42.000 699349.687 5669.020 36090.776 41759.797
43.000 662980.711 5390.821 36368.976 41759.797
44.000 626331.390 5110.476 36649.320 41759.797
45.000 589399.565 4827.971 36931.825 41759.797
46.000 552183.057 4543.288 37216.508 41759.797
47.000 514679.671 4256.411 37503.386 41759.797
48.000 476887.197 3967.322 37792.474 41759.797
49.000 438803.406 3676.005 38083.791 41759.797
50.000 400426.052 3382.443 38377.354 41759.797
51.000 361752.873 3086.617 38673.179 41759.797
52.000 322781.588 2788.512 38971.285 41759.797
53.000 283509.900 2488.108 39271.689 41759.797
54.000 243935.492 2185.389 39574.408 41759.797
55.000 204056.031 1880.336 39879.461 41759.797
56.000 163869.167 1572.932 40186.635 41759.797
57.000 123372.528 1263.158 40496.868 41759.797
58.000 82563.728 950.997 40808.800 41759.797
59.000 41440.360 636.429 41123.368 41759.797
60.000 0.000 319.436 41440.360 41759.797

```

```
(%i4) quit();
```

If you want to be a little adventurous in your monthly payments, that is, instead of equal payments, you want to do increasing payments, Maxima could help you with that as well. `arit_amortization()` and `geo_amortization()` are two such functions, which provide the schedule for increasing payments with fixed amount increments and fixed rate increments, respectively. Here's a small demo of how it works:

```

$ maxima -q
(%i1) load(finance)$
(%i2) amortization(0.10, 100, 5)$

"n"      "Balance"    "Interest"    "Amortization"  "Payment"
0.000    100.000    0.000        0.000          0.000
1.000    83.620     10.000       16.380         26.380
2.000    65.603     8.362        18.01          26.380
3.000    45.783     6.560        19.819         26.380
4.000    23.982     4.578        21.801         26.380
5.000    0.000      2.398        23.982         26.380

```

```

(%i3) arit_amortization(0.10, 10, 100, 5)$
"n"      "Balance"    "Interest"    "Amortization"  "Payment"

```

```

0.000    100.000    0.000        0.000          0.000
1.000    101.722    10.000       -1.722          8.278
2.000    93.615    10.172        8.106          18.278
3.000    74.698    9.362         18.917         28.278
4.000    43.890    7.470         30.809         38.278
5.000    0.000     4.389         43.890         48.278

```

```

(%i4) geo_amortization(0.10, 0.05, 100, 5)$
"n"      "Balance"    "Interest"    "Amortization"  "Payment"
0.000    100.000    0.000        0.000          0.000
1.000    85.907    10.000       14.093         24.093
2.000    69.200    8.591        16.707         25.298
3.000    49.558    6.920        19.642         26.562
4.000    26.623    4.956        22.935         27.891
5.000    0.000     2.662        26.623         29.285

```

```
(%i5) quit();
```

%i2 has been provided for comparative analysis. %i3 shows the incremental payout with increments of ₹10 (the second parameter of `arit_amortization()`). %i4 shows the incremental payout with increments at the rate of 5 per cent (the second parameter of `geo_amortization()`). Both these computations could be done in decrements as well – just pass the second parameter as negative.

Plan your savings

Say, you have a savings account like the PPF (Public Provident Fund), giving you interest at a rate of 8 per cent p.a., and at the end of five years, you want to have saved ₹100,000. So, what should be your minimum yearly deposit into your account? It is not as simple as dividing by 5, as the interest would be also added to your savings. `saving()` shows you the complete schedule as follows:

```

$ maxima -q
(%i1) load(finance)$
(%i2) saving(0.08 /* interest rate */, 100000 /* final savings */,
5 /* years */) $
"n"      "Balance"    "Interest"    "Payment"
0.000    0.000        0.000          0.000
1.000    17045.645    0.000         17045.645
2.000    35454.943    1363.652      17045.645
3.000    55336.983    2836.395      17045.645
4.000    76809.588    4426.959      17045.645
5.000    100000.000    6144.767      17045.645

```

And the minimum yearly deposit to be made is ₹17,046. The 'Balance' and 'Interest' columns, respectively, tell you about the balance and the interest accumulated in the corresponding year. If you are only interested in knowing the minimum amount to be deposited, you may just use the `annuity_fv()` function, which basically computes the annuity of ₹17,046 every year for five years in order to have saved ₹100,000 after five years.